

Exchange Rate Movements and Armenian Inflation: A Multiple Regression Analysis with Armenian CPI and FX Data

A. Pambukyan, T. Petrosyan, A. Musoyan

CS108: Statistics

American University of Armenia

Instructor: Dr. A. Abrahamyan

December 4, 2025

Abstract

The Armenian dram (AMD) is the currency of a small open economy where imports play an important role in consumption. It is therefore natural to ask whether movements in the AMD exchange rate help explain or forecast inflation. In this project we use monthly Armenian data to study the relationship between consumer price inflation and AMD exchange rates against the US dollar (USD), the euro (EUR), and the Russian ruble (RUB).

We construct monthly inflation as the log difference of the national consumer price index (CPI) published by ArmStat and compute monthly log changes of AMD/USD, AMD/EUR, and AMD/RUB based on daily exchange rates from the Central Bank of Armenia. Using multiple linear regression, we estimate a sequence of nested models: (i) monthly inflation on its own lag and seasonal dummies; (ii) the same model augmented by contemporaneous exchange rate changes; (iii) one-step ahead inflation on current inflation and exchange rate changes; and (iv) twelve-month cumulative inflation on twelve-month cumulative exchange rate changes, with and without lagged twelve-month inflation.

For each pair of nested models we use standard t tests and F tests to assess whether the block of exchange rate variables adds statistically significant explanatory power, and we complement the numerical results with residual diagnostics and outlier checks. At the monthly horizon we find only modest evidence for exchange rate effects. Given current inflation and seasonality, exchange rate changes do not significantly improve the prediction of next month inflation. At the twelve-month horizon, however, cumulative AMD depreciation against foreign currencies, in particular against the Russian ruble, is strongly associated with higher year-on-year inflation, even after controlling for the high persistence of inflation itself. We discuss the statistical and economic interpretation of these findings, as well as their limitations.

Code and data. All computations are performed in R. Scripts and cleaned data files are available upon request.

1 Introduction

Inflation is one of the key macroeconomic indicators for any economy. In a small open economy like Armenia, movements in the exchange rate of the national currency can affect domestic prices through the cost of imported goods and services. Understanding how much and on what horizon exchange rate changes are reflected in consumer prices is important for monetary policy, forecasting, and risk management.

The informal economic story is that if the Armenian dram depreciates, imports become more expensive in AMD, and at least part of this higher cost is likely to appear in the consumer price index (CPI). The statistical question is do exchange rate movements add useful information for explaining or predicting Armenian inflation?

In this project we address this question using standard regression tools. We work entirely with Armenian data and deliberately keep the methodology simple: multiple linear regression, t tests for individual coefficients, F tests for blocks of variables, residual analysis, and simple model comparisons.

Research questions

We focus on three related questions.

- (Q1) *Contemporaneous association.* Do monthly AMD exchange rate changes help explain monthly Armenian CPI inflation after controlling for past inflation and seasonality?
- (Q2) *Short horizon prediction.* Given information at month t , including current inflation and exchange rate changes, do exchange rates help forecast inflation at month $t + 1$?
- (Q3) *Medium horizon association.* Do twelve-month cumulative exchange rate changes help explain twelve-month cumulative inflation, and does this remain true after controlling for the strong persistence of inflation?

Our main empirical finding is that the answer is mostly “no” at the one-month horizon and clearly “yes” at the twelve-month horizon, especially for the AMD/RUB exchange rate.

2 Data and transformations

2.1 CPI and inflation

We use the official consumer price index (CPI) for the Republic of Armenia from the Arm-StatBank online database. Concretely, we work with the table “*Consumer price index in the Republic of Armenia, compared to the previous month (1993–2025)*” which provides monthly CPI indices at a monthly frequency.¹ The raw index at month t is denoted by C_t .

Following standard practice, we define monthly log inflation as

$$Y_t = 100(\log C_t - \log C_{t-1}), \quad t = 2, \dots, T, \quad (1)$$

so that Y_t is approximately the percentage change of the CPI between months $t - 1$ and t . We refer to Y_t simply as “monthly inflation”. In the R code this is the variable `infl_pct`.

2.2 Exchange rates

Exchange rates of the Armenian dram against foreign currencies are obtained from the Central Bank of Armenia (CBA) exchange rate archive, which reports daily official AMD rates against major currencies.² We focus on the AMD exchange rate against three currencies:

- US dollar (USD);
- euro (EUR);
- Russian ruble (RUB).

For each currency and each month we compute the average daily rate, then define the monthly log change

$$X_t^{\text{USD}} = 100(\log S_t^{\text{USD}} - \log S_{t-1}^{\text{USD}}), \quad (2)$$

¹Available at <https://statbank.armstat.am>, section 1.2.1 “Monthly indicators”.

²Exchange rate archive: <https://www.cba.am/en/exchange-rates-archive>.

where S_t^{USD} is the average AMD per USD rate in month t . Positive values of X_t^{USD} therefore correspond to AMD depreciation against the dollar. Analogous definitions give X_t^{EUR} and X_t^{RUB} . In the R code these are `dlog_usd`, `dlog_eur`, and `dlog_rub`.

The CPI and exchange rate data are merged by month into a single panel `data_all`. We restrict attention to the period where all three series overlap and the required lags are available.

2.3 Twelve-month cumulative variables

To study exchange rate pass-through at a longer horizon, we construct approximate twelve-month cumulative inflation and exchange rate changes by summing the last twelve monthly log changes. Formally,

$$Y_t^{(12)} = \sum_{j=0}^{11} Y_{t-j}, \quad X_t^{\text{USD}(12)} = \sum_{j=0}^{11} X_{t-j}^{\text{USD}}, \quad (3)$$

and similarly for EUR and RUB. In the R code these appear as `infl12`, `usd12`, `eur12`, `rub12`. This transformation turns our monthly data into a rolling twelve-month window and is the discrete analogue of year-on-year inflation.

2.4 Seasonality and notation

To account for systematic seasonal patterns, we include monthly dummy variables. Let $D_{m,t}$ be an indicator equal to 1 if month t is calendar month m and 0 otherwise, for $m = 1, \dots, 12$. We omit one month to avoid perfect collinearity. In the code, these dummies are encoded via a factor variable `month_factor`.

Throughout, we write Y_t for monthly inflation, $Y_t^{(12)}$ for twelve-month cumulative inflation, and use superscripts for the different currencies.

3 Statistical models

This section formulates the regression models we estimate. All models are linear and estimated by ordinary least squares (OLS) on the available time range.

Even though the true data-generating process for inflation and exchange rates is likely more complex and may involve nonlinearities and richer dynamics, a linear specification is still a reasonable choice in our setting. It provides a first-order approximation to average relationships, keeps the coefficients easy to interpret in terms of marginal effects, and matches the level of methods covered in this course. Moreover, the residual and influence diagnostics in Section 5 suggest that, for our sample, the linear model captures the main systematic patterns in the data reasonably well.

3.1 Monthly inflation: baseline and FX-augmented

Our first models address Question (Q1). We write Y_t for monthly inflation and Y_{t-1} for its one-month lag. The baseline model controls only for persistence and seasonality:

$$(M1) \quad Y_t = \beta_0 + \beta_1 Y_{t-1} + \sum_{m=1}^{11} \gamma_m D_{m,t} + \varepsilon_t, \quad (4)$$

where December is the omitted month. The error term ε_t is assumed to have mean zero and constant variance, and we treat the model as an approximation to the true data-generating process.

To test whether exchange rate changes help explain monthly inflation, we augment the model by contemporaneous FX changes:

$$(M2) \quad Y_t = \beta_0 + \beta_1 Y_{t-1} + \sum_{m=1}^{11} \gamma_m D_{m,t} + \delta_1 X_t^{\text{USD}} + \delta_2 X_t^{\text{EUR}} + \delta_3 X_t^{\text{RUB}} + \varepsilon_t. \quad (5)$$

Model (M1) is nested inside (M2). We use:

- t tests to assess the individual significance of each δ_j ;
- an F test to test the joint null hypothesis

$$H_0 : \delta_1 = \delta_2 = \delta_3 = 0$$

against the alternative that at least one is nonzero.

We also compare R^2 and adjusted R^2 between (M1) and (M2).

3.2 Predicting next-month inflation

To address Question (Q2), we construct $\tilde{Y}_t = Y_{t+1}$, the inflation one month ahead. The baseline model uses only current inflation and seasonality:

$$(N1) \quad \tilde{Y}_t = \alpha_0 + \alpha_1 Y_t + \sum_{m=1}^{11} \eta_m D_{m,t} + u_t, \quad (6)$$

while the FX-augmented model adds current exchange rate changes:

$$(N2) \quad \tilde{Y}_t = \alpha_0 + \alpha_1 Y_t + \sum_{m=1}^{11} \eta_m D_{m,t} + \theta_1 X_t^{\text{USD}} + \theta_2 X_t^{\text{EUR}} + \theta_3 X_t^{\text{RUB}} + u_t. \quad (7)$$

Again, we use t tests for individual coefficients and an F test for the block $\theta_1, \theta_2, \theta_3$, as well as a comparison of out-of-sample fit in a simple training-test split if time allows.

3.3 Twelve-month inflation

To address Question (Q3), we work with the twelve-month cumulative variables $Y_t^{(12)}$ and $X_t^{(12)}$ defined earlier. Our first model regresses twelve-month inflation on seasonal dummies only:

$$(Y1) \quad Y_t^{(12)} = \beta_0^{(12)} + \sum_{m=1}^{11} \gamma_m^{(12)} D_{m,t} + e_t. \quad (8)$$

The FX-augmented version adds twelve-month exchange rate changes:

$$(Y2) \quad Y_t^{(12)} = \beta_0^{(12)} + \sum_{m=1}^{11} \gamma_m^{(12)} D_{m,t} + \delta_1^{(12)} X_t^{\text{USD}(12)} + \delta_2^{(12)} X_t^{\text{EUR}(12)} + \delta_3^{(12)} X_t^{\text{RUB}(12)} + e_t. \quad (9)$$

Finally, to account for the high persistence of twelve-month inflation, we add a lag term:

$$(Y3) \quad Y_t^{(12)} = \phi_0 + \phi_1 Y_{t-1}^{(12)} + \sum_{m=1}^{11} \kappa_m D_{m,t} + \xi_t, \quad (10)$$

and its FX-augmented version

$$(Y4) \quad Y_t^{(12)} = \phi_0 + \phi_1 Y_{t-1}^{(12)} + \sum_{m=1}^{11} \kappa_m D_{m,t} + \lambda_1 X_t^{\text{USD}(12)} + \lambda_2 X_t^{\text{EUR}(12)} + \lambda_3 X_t^{\text{RUB}(12)} + \xi_t. \quad (11)$$

Model (Y3) is nested inside (Y4). The coefficient ϕ_1 captures persistence. The joint significance of the FX block is again tested via an F test.

4 Empirical results

In this section we summarize the main empirical outputs from the R code and interpret them in statistical and economic terms.

4.1 Exploratory analysis

Figure 1 shows time plots of monthly inflation and exchange rate changes. Armenian inflation exhibits several spikes, including episodes of high inflation and deflation in the 1990s, followed by more moderate fluctuations in later years. Exchange rates also display periods of large AMD depreciation and appreciation, especially during global financial stress.

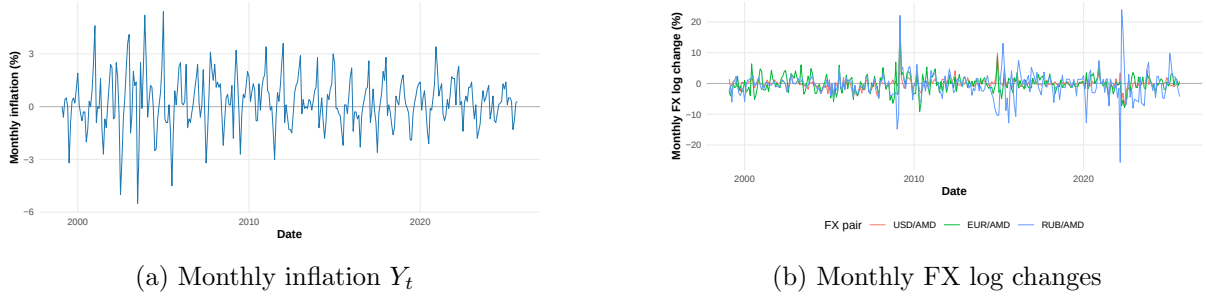


Figure 1: Time series of Armenian inflation and exchange rate changes.

Boxplots (not shown here) and simple summaries confirm the presence of several extreme observations in both inflation and exchange rate changes, corresponding to crisis periods. Because these are genuine economic events, we keep them in the main regressions and handle them through robustness checks rather than by deleting them from the sample.

4.2 Monthly inflation

Table 1 reports the results for the monthly models (M1) and (M2). In the baseline model (M1) the lagged inflation coefficient β_1 is about 0.27 (standard error 0.06, $p < 0.001$), indicating meaningful persistence in monthly inflation. Seasonal dummies capture small but noticeable differences across months and are jointly highly significant (not shown). The adjusted R^2 of (M1) is about 0.64.

Table 1: Monthly inflation: baseline (M1) and FX-augmented (M2) models.

	Estimate	Std. Error	t value	$\Pr(> t)$
Baseline model (M1): Y_t on Y_{t-1} and month dummies				
Intercept	1.960	0.203	9.647	< 0.001
Y_{t-1}	0.275	0.055	5.026	< 0.001
Month dummies (Feb–Dec)	included			
FX-augmented model (M2): add X_t^{USD} , X_t^{EUR} , X_t^{RUB}				
Intercept	1.904	0.204	9.313	< 0.001
Y_{t-1}	0.287	0.055	5.185	< 0.001
X_t^{USD}	0.020	0.036	0.550	0.583
X_t^{EUR}	0.049	0.023	2.101	0.036
X_t^{RUB}	0.001	0.012	0.055	0.956
Month dummies (Feb–Dec)	included			

The exchange rate coefficients in (M2) are small in magnitude. AMD/EUR changes have a modest positive coefficient (about 0.05) that is statistically significant at the 5 percent level, while AMD/USD and AMD/RUB changes are essentially insignificant. The F test that compares (M1) and (M2) rejects the null hypothesis that all three FX coefficients are zero at the 5 percent level ($F(3, 304) = 3.50$, $p = 0.016$), but the increase in adjusted R^2 is small (from 0.64 to 0.65). Taken together, these results suggest that exchange rate changes have at most a modest contemporaneous association with monthly inflation in this specification.

4.3 Next-month inflation

Table 2 summarizes the results for models (N1) and (N2), where the dependent variable is next-month inflation $\tilde{Y}_t = Y_{t+1}$. The baseline specification shows that current inflation is a significant predictor of next-month inflation, with an estimated coefficient around 0.27 and adjusted R^2 again around 0.64.

Table 2: Next-month inflation: baseline (N1) and FX-augmented (N2) models.

	Estimate	Std. Error	t value	$\Pr(> t)$
Baseline model (N1): Y_{t+1} on Y_t and month dummies				
Intercept	-1.000	0.221	-4.515	< 0.001
Y_t	0.275	0.055	5.007	< 0.001
Month dummies (Feb–Dec)	included			
FX-augmented model (N2): add X_t^{USD} , X_t^{EUR} , X_t^{RUB}				
Intercept	-0.977	0.222	-4.398	< 0.001
Y_t	0.260	0.056	4.686	< 0.001
X_t^{USD}	0.047	0.036	1.322	0.187
X_t^{EUR}	0.012	0.024	0.503	0.615
X_t^{RUB}	0.007	0.012	0.566	0.572
Month dummies (Feb–Dec)	included			

In these models the block of exchange rate variables is not statistically significant. The F test comparing (N1) and (N2) yields $F(3, 303) = 1.79$ with $p = 0.15$, and the adjusted R^2 changes only from about 0.64 to 0.64 (rounded). Intuitively, given current inflation and seasonal effects, current exchange rate movements do not materially improve the prediction of next-month inflation in this simple linear setting.

4.4 Twelve-month inflation

The twelve-month models provide the clearest evidence for exchange rate effects. Table 3 reports the results for models (Y1) and (Y2), where the dependent variable is twelve-month cumulative inflation $Y_t^{(12)}$.

In the FX-augmented model (Y2) the block of twelve-month exchange rate changes is strongly statistically significant. The F test comparing (Y1) and (Y2) yields $F(3, 294) = 11.9$ with $p < 0.001$, and the adjusted R^2 increases from essentially zero in (Y1) to about 0.07 in (Y2). The coefficient on twelve-month USD depreciation is negative and significant, while the coefficient on twelve-month RUB depreciation is positive and highly significant; the EUR coefficient is small and insignificant. At least one of the currencies therefore shows a clear association with twelve-month inflation, and in our sample the AMD/RUB variable plays a particularly important role.

To control more directly for persistence, we turn to models (Y3) and (Y4), which include lagged twelve-month inflation $Y_{t-1}^{(12)}$. The lag coefficient ϕ_1 is large and highly significant,

Table 3: Twelve-month inflation: baseline (Y1) and FX-augmented (Y2) models.

	Estimate	Std. Error	t value	$\Pr(> t)$
Baseline model (Y1): $Y_t^{(12)}$ on month dummies				
Intercept	3.542	0.658	5.380	< 0.001
Month dummies (Feb–Dec)		included		
FX-augmented model (Y2): add $X^{(12)}$ for USD, EUR, RUB				
Intercept	3.828	0.632	6.059	< 0.001
$X_t^{\text{USD}(12)}$	-0.104	0.025	-4.217	< 0.001
$X_t^{\text{EUR}(12)}$	0.027	0.021	1.304	0.193
$X_t^{\text{RUB}(12)}$	0.053	0.014	3.741	< 0.001
Month dummies (Feb–Dec)		included		

reflecting the slow movement of year-on-year inflation. Table 4 summarizes these results.

Table 4: Twelve-month inflation with lag and FX: models (Y3) and (Y4).

	Estimate	Std. Error	t value	$\Pr(> t)$
Baseline model (Y3): $Y_t^{(12)}$ on $Y_{t-1}^{(12)}$ and month dummies				
Intercept	0.189	0.231	0.821	0.412
$Y_{t-1}^{(12)}$	0.943	0.019	49.713	< 0.001
Month dummies (Feb–Dec)		included		
FX-augmented model (Y4): add $X^{(12)}$ for USD, EUR, RUB				
Intercept	0.294	0.229	1.282	0.201
$Y_{t-1}^{(12)}$	0.942	0.020	47.924	< 0.001
$X_t^{\text{USD}(12)}$	0.006	0.009	0.641	0.522
$X_t^{\text{EUR}(12)}$	0.008	0.007	1.159	0.248
$X_t^{\text{RUB}(12)}$	0.013	0.005	2.790	0.006
Month dummies (Feb–Dec)		included		

Two points stand out.

- The coefficient on lagged twelve-month inflation is close to one and highly significant in both models, which is consistent with the idea that year-on-year inflation changes slowly (adjusted R^2 is about 0.89 in (Y3)).
- Even after controlling for this strong persistence, the FX block remains jointly significant: the F test comparing (Y3) and (Y4) gives $F(3, 292) = 5.65$ with $p < 0.001$, and the adjusted R^2 rises slightly to about 0.89 in (Y4). The twelve-month AMD/RUB change is individually significant at conventional levels, with a positive coefficient. A cumulative depreciation of the dram against the ruble is associated with higher twelve-month inflation in Armenia.

These results are broadly robust, in the sense that they are driven by a pattern across many months rather than a single data point. The Cook’s distance analysis below shows only a handful of moderately influential observations, mostly corresponding to large exchange rate movements during crisis periods, so trimming a few extreme RUB observations would be expected to change the numerical values somewhat without overturning the qualitative pattern.

5 Diagnostics and robustness

Regression diagnostics are important for checking whether the linear model assumptions are at least approximately reasonable. We focus on the final twelve-month model (Y4), since it is our main specification.

5.1 Residual analysis

Figure 2 shows residual plots for model (Y4): residuals versus fitted values, residuals over time, a normal QQ plot, and a histogram of residuals.

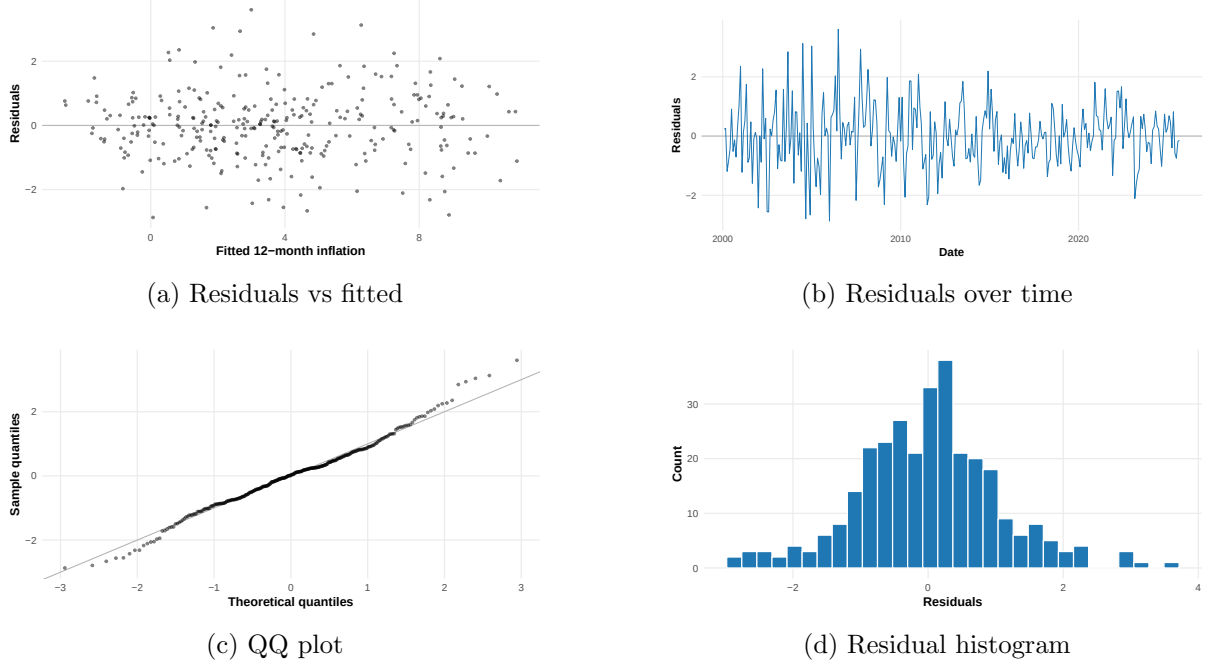


Figure 2: Residual diagnostics for the twelve-month FX-augmented model (Y4).

The residual versus fitted plot does not show a strong systematic pattern, although some heteroskedasticity is possible at extreme fitted values. Residuals over time indicate that the model captures most of the slow movement in twelve-month inflation, with remaining deviations that are relatively small except during crisis periods. The QQ plot and histogram suggest mild deviations from normality in the tails, which is not surprising for macroeconomic data, but the bulk of the distribution is reasonably symmetric.

5.2 Influence and outliers

To assess the influence of individual observations, we compute Cook's distance for model (Y4). Figure 3 plots Cook's distance over time and marks a common rule-of-thumb threshold at $4/n$, where n is the sample size.

A small number of months show relatively high influence, mostly corresponding to large exchange rate movements during crisis periods. Because the influential points are few, they mainly affect the precise size of the FX coefficients rather than their sign; the qualitative conclusion that twelve-month AMD/RUB movements matter for twelve-month inflation is unlikely to depend on a single month.

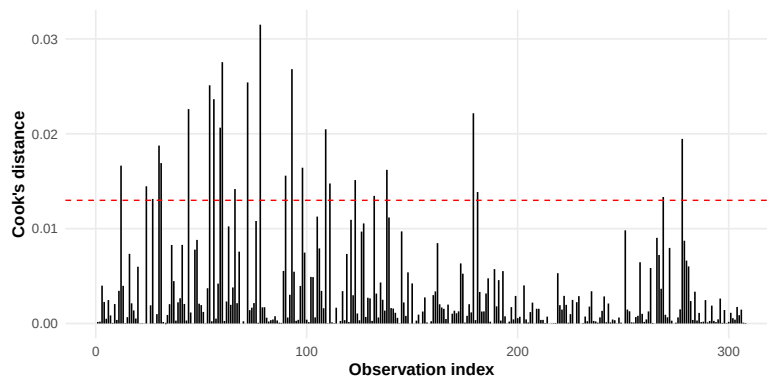


Figure 3: Cook's distance for the twelve-month FX-augmented model (Y4).

6 Discussion and conclusion

This project studied the relationship between Armenian consumer price inflation and AMD exchange rates against USD, EUR and RUB using only tools from a standard undergraduate regression course.

At the monthly horizon, we find limited evidence that exchange rate changes add explanatory power for inflation once we control for lagged inflation and seasonal dummies. Even when the block of FX variables is jointly significant at the 5 percent level, the improvement in adjusted R^2 is small, and the economic magnitude of the estimated coefficients is modest.

For one-month ahead prediction, current exchange rate changes do not significantly improve the forecast of next-month inflation, given current inflation and seasonality. This is consistent with the idea that much of the exchange rate information is either already incorporated into current prices or is not transmitted to consumer prices on such a short horizon in a linear way.

The main result concerns twelve-month cumulative inflation. When we aggregate both inflation and exchange rate changes over a year, twelve-month FX changes clearly help explain twelve-month inflation. This is especially true for AMD/RUB changes, which remain individually significant even after controlling for the strong persistence of twelve-month inflation. Our interpretation is that medium-term exchange rate movements against key trading partners, in particular Russia, are an important part of the story for Armenian inflation dynamics.

From a statistical perspective, the project illustrates how far one can go with simple multiple regression, careful model formulation, and standard diagnostics. All hypothesis tests are standard t and F tests, and all models are linear. At the same time, the time series context reminds us that independence assumptions are only approximate, and that lagged variables and cumulative sums are often needed to capture persistence.

There are several limitations. The models are purely reduced-form and omit other macro variables such as world commodity prices, output gaps or monetary policy indicators. We use simple linear specifications and do not model volatility or structural breaks. Nevertheless, within the scope of the course, our analysis provides a statistically grounded view of exchange rate pass-through in Armenia and shows that the effect is more visible at a twelve-month horizon than at the monthly level.

Future work could extend the analysis by including additional macro variables, exploring non-linear specifications or regime changes, or using more advanced time series methods such as vector autoregressions or state space models.

References

- [1] Statistical Committee of the Republic of Armenia. “Consumer price index in the Republic of Armenia, compared to the previous month (1993–2025)”, ArmStatBank online database. Available at https://statbank.armstat.am/pxweb/en/ArmStatBank/ArmStatBank__1%20Economy%20and%20finance__12%20Consumer%20Prices__1.2.1%20Monthly%20indicators/EF-cpi-mi1.px/table/tableViewLayout1/.
- [2] Central Bank of Armenia. “Exchange rates archive”, official website of the Central Bank of Armenia. Available at <https://www.cba.am/en/statistics/exchange-rate/157/>.